
MAKERERE UNIVERSITY
Department of Computer Science
CSC 1105: Mathematics for Computer Science

ASSIGNMENT 1
Limits, Continuity and Derivatives

Instructions

1. This assignment is due on **September 1, 2026**.
2. Students should work in groups of a maximum of **six (6) members**.
3. The assignment must be **typed, printed, and submitted in hard copy**.
4. Attempt all questions.
5. Show all essential mathematical steps. Answers without sufficient justification may receive little or no credit.
6. Where a particular theorem or limit technique is used, clearly state it.
7. Groups will be required to present their solutions in class or online.

Question 1

Consider the function

$$f(x) = \frac{x^2 - 4x + 3}{x - 1}, \quad x \neq 1.$$

- (a) Determine the domain of f .
- (b) Construct a table of values of $f(x)$ using

$$x = 0.9, 0.99, 0.999, 1.001, 1.01, 1.1.$$

Hence estimate

$$\lim_{x \rightarrow 1} f(x).$$

- (c) Simplify $f(x)$ for $x \neq 1$ and use the simplified expression to verify your answer in part (b).
- (d) Explain why the existence of $\lim_{x \rightarrow 1} f(x)$ does not require $f(1)$ to be defined.

Question 2

A computer system changes its resource-allocation rule when a load index x reaches 4. Its response score is modeled by

$$R(x) = \begin{cases} 2x + 1, & x < 4, \\ 12, & x = 4, \\ x^2 - 7, & x > 4. \end{cases}$$

(a) Determine

$$\lim_{x \rightarrow 4^-} R(x) \quad \text{and} \quad \lim_{x \rightarrow 4^+} R(x).$$

(b) Does $\lim_{x \rightarrow 4} R(x)$ exist? Give a mathematical justification.

(c) Determine $R(4)$ and explain whether $R(4)$ affects the existence of the two-sided limit.

(d) Sketch the graph of $R(x)$ in a neighbourhood of $x = 4$, clearly distinguishing open and closed points.

(e) State the condition under which a two-sided limit at a point exists.

Question 3

Suppose

$$\lim_{x \rightarrow 2} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 2} g(x) = -2.$$

Without knowing explicit formulas for f and g , evaluate the following, where possible:

(a)

$$\lim_{x \rightarrow 2} [4f(x) - 3g(x)].$$

(b)

$$\lim_{x \rightarrow 2} f(x)g(x).$$

(c)

$$\lim_{x \rightarrow 2} \frac{f(x) + g(x)}{f(x) - g(x)}.$$

(d) Using direct substitution, evaluate

$$\lim_{x \rightarrow -2} \frac{2x^3 - 3x^2 + x + 6}{x^2 + 5}.$$

State why direct substitution is valid.

Question 4

Evaluate each of the following limits. For each one, explicitly state the main technique used.

(a)

$$\lim_{x \rightarrow 5} \frac{x^2 - 7x + 10}{x - 5}.$$

(b)

$$\lim_{x \rightarrow 0} \frac{\sqrt{x + 16} - 4}{x}.$$

(c)

$$\lim_{x \rightarrow 0} \frac{\sin(6x)}{\sin(2x)}.$$

(d) Use the Squeeze Theorem to evaluate

$$\lim_{x \rightarrow 0} x^3 \cos\left(\frac{5}{x}\right).$$

Your solution must explicitly identify suitable lower and upper bounding functions.

Question 5

(a) Before applying L'Hopital's Rule, determine the indeterminate form of

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{e^{2x} - 1}.$$

(b) Hence evaluate the limit using L'Hopital's Rule.

(c) Evaluate

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}.$$

You may apply L'Hopital's Rule more than once, but every application must be justified.

(d) A student applies L'Hopital's Rule directly to

$$\lim_{x \rightarrow 0} \frac{x^2 + 1}{x + 1}.$$

Explain why this is inappropriate and evaluate the limit correctly.

Question 6

A simplified performance model for a server is

$$P(n) = \frac{120n^2 + 15n + 8}{4n^2 + 25},$$

where n represents the number of simultaneous requests, measured in hundreds.

(a) Evaluate

$$\lim_{n \rightarrow \infty} P(n).$$

(b) Identify the horizontal asymptote of P .

(c) Interpret the limit in the context of server performance.

(d) Evaluate

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}.$$

(e) Evaluate

$$\lim_{x \rightarrow \infty} x^2 e^{-x}.$$

You may use L'Hopital's Rule after rewriting the expression as an appropriate quotient.

Question 7

(a) Consider

$$f(x) = \begin{cases} ax + 2, & x < 3, \\ 11, & x = 3, \\ x^2 + b, & x > 3. \end{cases}$$

Determine the values of a and b for which f is continuous at $x = 3$.

(b) Classify the discontinuity of each function at the indicated point as **removable**, **jump**, or **infinite**. Justify each classification.

(i)

$$p(x) = \frac{x^2 - 9}{x - 3}, \quad x = 3.$$

(ii)

$$q(x) = \begin{cases} x + 1, & x < 2, \\ 3x - 1, & x \geq 2, \end{cases} \quad x = 2.$$

(iii)

$$r(x) = \frac{1}{(x - 5)^2}, \quad x = 5.$$

(c) State the three mathematical conditions required for a function to be continuous at $x = a$.

Question 8

A software engineer uses the function

$$F(x) = x^3 - 5x - 3$$

in a numerical root-finding routine.

- (a) Show that F is continuous on the interval $[2, 3]$.
- (b) Compute $F(2)$ and $F(3)$ and use the Intermediate Value Theorem to prove that the equation

$$x^3 - 5x - 3 = 0$$

has at least one root in $(2, 3)$.

- (c) Without solving the equation exactly, determine a smaller interval of length 0.5 that contains a root. Justify your answer using the IVT.

Question 9

Consider

$$f(x) = |2x - 4|.$$

- (a) Write $f(x)$ as a piecewise-defined function.
- (b) Show using one-sided limits that f is continuous at $x = 2$.
- (c) Determine the left-hand and right-hand derivatives at $x = 2$. Hence determine whether f is differentiable at $x = 2$.
- (d) Explain what this example demonstrates about the relationship between continuity and differentiability.
- (e) Using the definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

derive from first principles the derivative of

$$g(x) = x^2 + 3x.$$

Question 10

Differentiate each of the following functions. Simplify your answers where appropriate.

- (a) $y = 5x^7 - 4x^3 + 6x - 9$.

(b) $y = (x^2 + 3x)(2x^3 - 5)$.

(c) $y = (4x^3 - 2x + 7)^5$.

Question 11

During testing of a mobile data service in Kampala, the cumulative amount of data transmitted by a test server after t seconds is modeled by

$$D(t) = 2t^3 - 9t^2 + 24t,$$

where D is measured in megabytes.

- (a) Find an expression for the instantaneous data transmission rate $D'(t)$.
- (b) Determine the instantaneous transmission rate at $t = 3$ seconds. Include appropriate units.
- (c) Find the acceleration of the transmission rate, that is, $D''(t)$, and evaluate it at $t = 3$.
- (d) Interpret the sign of $D''(3)$ in the context of the system.

Question 12

A Ugandan technology company is configuring the number x of processing units allocated to a machine-learning task. A simplified model for the computational cost is

$$C(x) = 2x^2 - 32x + 180,$$

where $x > 0$ and $C(x)$ is measured in abstract cost units.

- (a) Find $C'(x)$ and determine the critical value of x .
- (b) Use the second derivative to determine whether this critical value gives a minimum or maximum.
- (c) Determine the minimum computational cost.
- (d) Explain briefly why derivatives are useful when optimizing computational or machine-learning systems.

End